STUDIO 5000 STRUCTURED TEXT

Tag name is DWELL *using Data Type* **FBD_TIMER**.

Timer structure

```
DWELL.PRE := 5000;
DWELL.TimerEnable := SWITCH;
TONR(DWELL);
```

Format when dragged in:

```
TONR( )
```

Timer status bit formats

```
DWELL.EN
DWELL.TT
DWELL.DN
```

Examples:

Energize Timer Delay ON

```
DWELL.PRE := 5000;
DWELL.TimerEnable:= SWITCH;
TONR(DWELL);
OUTPUT := DWELL.DN;
```

Energize Timer Delay OFF

```
DWELL.PRE := 5000;
DWELL.TimerEnable:= SWITCH;
TONR(DWELL);
OUTPUT := NOT DWELL.DN AND DWELL.EN;
```

Setting Preset

```
DWELL.PRE  := TIME_VALUE;
```

Energize Timer Delay ON

```
DWELL.PRE := 5000;
DWELL.TimerEnable := SWITCH;
TONR(DWELL);
IF DWELL.DN THEN
     OUTPUT := 1;
ELSE;
     OUTPUT := 0;
END_IF;
```

Energize Timer Delay OFF

```
DWELL.PRE:= 5000;
DWELL.TimerEnable := SWITCH;
TONR(DWELL);
IF NOT DWELL.DN AND DWELL.EN THEN
     OUTPUT := 1;
ELSE;
     OUTPUT := 0;
END_IF;
```

Tag name ONE_SHOT *using Data Type* **FBD_ONESHOT**.

One Shot Rising structure

```
ONE_SHOT.InputBit := PUSHBUTTON;
OSRI(ONE_SHOT);
```

Format when dragged in:

```
OSRI( )
```

One shot bit (the tag for the one scan pulse)

```
IF ONE_SHOT.OutputBit AND OK_TO_RUN THEN
      MY_TAG := 1;
ELSE
      MY_TAG := 0;
END_IF;
```

Format for assigning Tag to Output Bit:

```
ONE_SHOT.InputBit := PUSHBUTTON;
OSRI(ONE_SHOT);
PULSE := ONE_SHOT.OutputBit
```

SIEMENS STRUCTURED TEXT

A dialog opens when the TON *instruction is dragged into the program, the tag name in this example is* DWELL. *The* TON *is a System block and Program resource.*

Timer structure

```
“DWELL".TON(IN:= "SWITCH",
          PT:=T#5000ms);
```

Timer status bit formats

```
"DWELL".Q
"DWELL".IN
```

Format when dragged in:

```
"DWELL".TON(IN:=_bool_in_,
            PT:=_time_in_,
            Q=>_bool_out_,
            ET=>_time_out_);
```

The IN and PT are required.

If the Q and ET are unassigned, they will be deleted when press ENTER to complete the instruction.

Examples:

Energize Timer Delay ON

```
“DWELL".TON(IN:= "SWITCH",
            PT:=T#5000ms);
"OUTPUT" := NOT "DWELL".Q;
```

Energize Timer Delay OFF

```
“DWELL".TON(IN:= "SWITCH",
            PT:=T#5000ms);
"OUTPUT" := NOT "DWELL".Q AND "DWELL".IN;
```

Setting Prest Time

```
"DWELL".PT := T#5000ms;
```

Energize Timer Delay ON

```
“DWELL".TON(IN:= "SWITCH",
            PT:=T#5000ms);
IF "DWELL".Q THEN
   "OUTPUT" := 1;
ELSE;
   "OUTPUT" := 0;
END_IF;
```

Energize Timer Delay OFF

```
“DWELL".TON(IN:= "SWITCH",
           PT:=T#5000ms);
IF NOT "DWELL".Q THEN
   "OUTPUT" := 1;
ELSE;
   "OUTPUT" := 0;
END_IF;
```

A dialog opens when the R_TRIG *instruction is dragged into the program, the tag name in this example is* ONE_SHOT. *The* R_TRIG *is a System block and Program resource.*

One Shot Rising structure

```
“ONE_SHOT"(CLK:="PUSHBUTTON",
                  Q=>#PULSE);
        (#PULSE is a Local Instance)
```

One shot bit (the tag for the one scan pulse)

```
IF #PULSE AND OK_TO_RUN THEN
    MY_TAG := 1;
ELSE
    MY_TAG := 0;
END_IF;
```

Format when dragged in:

```
“ONE_SHOT”(CLK:=_bool_in,
            Q=>_bool_out_);
```

Notice the instruction name is not shown.